



KELLEY SCHOOL OF BUSINESS

INDIANA UNIVERSITY

Doctoral Programs

Operations and Decision Technologies Information Systems Major

The Ph.D. program in Information Systems (IS) prepares students for careers as academic scholars and researchers focused on the design, use, and impact of business technologies. Research in the program examines how information technologies enable and empower individuals, teams, business processes, organizations, and society. Doctoral students work closely with faculty whose expertise spans a broad range of research areas, including artificial intelligence, IT implementation, IT-enabled decision-making, information systems design, information goods, healthcare IT, conversational AI, social media, collaborative work, usability, data analytics, recommender systems, crowdsourcing and crowdfunding, cybersecurity, mental health informatics, and related topics.

The program offers substantial flexibility, allowing students to take courses both within and outside the department to develop expertise aligned with their research interests. Students are encouraged to begin research early in the program, and several recent graduates have published their work in leading academic journals before entering the job market.

Plan of Study

The Ph.D. in Information Systems is typically completed in four to five years. During the first two years, students focus primarily on coursework and research development. Depending on their academic background, students may take selected MBA-level courses in addition to doctoral-level coursework in information systems, data analysis, and research methods. Students complement their IS coursework with methodological training in areas such as research methods, statistics, econometrics, and structural equation modeling. They also complete coursework in a minor area that supports their individual research interests. Possible minors include, but are not limited to, management, business economics, psychology, marketing, data science, and informatics.

For more information on the Information Systems major, contact: Professor Antino Kim at antino@iu.edu

Primary Doctoral Faculty

Hillol Bala, Conrad Prebys Professor of Information Systems; Ph.D., University of Arkansas, 2008. Research interests: digitalization (e.g., AI, digital platforms, and modern enterprise systems) in organizations and society, including healthcare contexts; IT-enabled organizational change; IT use, adaptation, and impact.

Naveen Basavaraj, Assistant Professor; Ph.D., Carnegie Mellon University, 2025. Research interests: Technology-driven societal changes.

A. Benedikt Brendel, Assistant Professor; Dr., University of Goettingen, 2017. Research interests: customer service failure and recovery, humanlike AI interfaces (e.g., chatbots), Human-AI collaboration, and affective reactions to AI (e.g., fear).

Nick Brown, Assistant Professor; Ph.D., Virginia Tech, 2023. Research interests: Information privacy, information security, and market manipulations (e.g., dark patterns, misinformation)

Kanghyun (Simon) Cho, Assistant Professor; Ph.D., Temple University. 2026. Research interests: Future of Work, Human-AI collaboration, AI in healthcare, Impact of Health IT, and Fintech.

Alan Dennis, Distinguished Professor of Information Systems and John T. Chambers Chair of Internet Systems; Ph.D., University of Arizona, 1991. Research interests: conversational AI (chatbots and digital humans), AI agents, AI team members, and fake news on social media.

TJ Guo, Assistant Professor; Ph.D., University of Texas at Austin. 2025. Research interests: AI in health, Impact of Health IT.

Fujie Jin, Associate Professor; Ph.D., University of Pennsylvania, 2016. Research interests: impact of IT on organizations, digital economy, social media and crowdfunding.

Rain Kan, Assistant Professor; Ph.D., University of Washington, 2024. Research interests: digital transformation, human-AI interactions, online inclusivity.

Antino Kim, Associate Professor and Institute for Entrepreneurship & Competitive Enterprise (IECE) Faculty Fellow; Ph.D., University of Washington, 2016. Research interests: human-AI interactions, misinformation and social media, digital piracy, markets for information goods.

Sagar Samtani, Associate Professor and Sungkyunkwan Professor; Executive Founding Director, Data Science and Artificial Intelligence Lab (DSAIL); Ph.D., University of Arizona, 2018. Research interests: artificial intelligence, cybersecurity, cyber threat intelligence, deep learning, large language models, vulnerability management, Internet of Things, Dark Web analytics, mobile mental health, business intelligence.

Ankit Shah, Assistant Professor; Ph.D., George Mason University, 2019. Research interests: AI for cybersecurity, security of AI systems, AI-enabled decision optimization and analytics (reinforcement learning, large language models, computer vision, combinatorial optimization).

Siddhartha Sharma, Associate Professor; Ph.D., Carnegie Mellon University, 2020. Research interests: strategies and implications of digital platforms.

Ramesh Venkataraman, Professor of Information Systems and John R. Gibbs Professor; Ph.D., University of Arizona, 1995. Research interests: data modeling, heterogeneous databases, virtual teams and groupware, usability in mobile systems, software engineering, and database design.

Lei Wang, Assistant Professor; Ph.D., University of Connecticut, 2014. Research interests: artificial intelligence (AI), digital platforms, digital commerce, gamification, and user engagement.

Brad Wheeler, Sungkyunkwan Professor and IU James H. Rudy Professor; Ph.D., Indiana University, 1993. Research interests: Applied cybersecurity, CIO executive leadership, executive leadership of IT strategy, executive programs in business-IT effectiveness, net-enabled organizations, open-source software.

Lucy Yan, Professor and John & Esther Reese Professor; Ph.D., University of Washington, 2012. Research interests: social networks and large-scale data analysis, social media in healthcare, business analytics and data-driven decision making, economics of IS, mobile/internet marketing.

Tianyang Xie, Assistant Professor; Ph.D., University of Arizona, 2026. Research interests: machine learning, deep learning, causal inference, and AI simulation inference.

Jingjing Zhang, Professor of Information Systems and Judith Norman Davis and Kim G. Davis Professor of Business Analytics; Ph.D., University of Minnesota, 2012. Research interests: personalization and recommender systems, knowledge discovery and data mining, human-AI interactions.

Sample of Past Student Initial Placements

University of Texas, Austin (Aijia Yuan, 2026)
Auburn University (Ben Lazarine, 2026)
University Of Cincinnati (Warren Rosengren, 2026)
Ohio State University (Xiaotong Sun, 2026; Lecturer)
University of Arizona (Agrim Sachdeva, 2024)
University of Georgia (Akshat Lakhiwal, 2023)
University of Warwick (Seyoung Seol, 2021)
University of Delaware (Xiaoye Cheng, 2020)
University of Texas, Austin (Patricia Moravec, 2019)
University of Oklahoma (Xuan Feng, 2019)

Ranking Information

(June 17, 2026) [Kelley School ranked No. 1 in Association for Information Systems Research Rankings](#)

View Kelley's other rankings here: <https://kelley.iu.edu/about/rankings.html>

Other Resources

- FAQ from Kelley School of Business Doctoral Program Office:
<https://kelley.iu.edu/programs/phd/admissions/frequently-asked-questions/index.html>
- How to apply: <https://kelley.iu.edu/programs/phd/admissions/how-apply/index.html>

For more information on the Information Systems Major at the Kelley School of Business Doctoral Program, see: <https://kelley.iu.edu/faculty-research/departments/operations-decision-technologies/phd/information-systems.html>

General Information

- Graduates of the IS Ph.D. program pursue careers such as:
 - Faculty positions at business schools and other academic institutions
 - Research positions in industry, including organizations such as Amazon, Google, Microsoft, and other technology-focused companies

- Program overview

Years	Activities
1 – 2	Coursework Research assistant for faculty Summer papers Qualifying exam
3 – 4	Dissertation research Teach 2 courses Defend dissertation proposal
4 – 5	Job market Final dissertation defense

Note: Students are generally expected to complete the program within four to five years. The specific timeline varies depending on each student's research progress, career plans, and area of specialization. Students should consult with their current or prospective dissertation committee chair as early as possible to develop an appropriate plan.

- Coursework
 - Major: 18 credits
 - Minor: 9–12 credits
 - Students are encouraged to tailor the minor to reflect their individual research interests.
 - Methods & Analysis: 12 credits
 - Teaching Development: 1.5 credits
- Research environment
 - Students participate in an active research community that includes regular seminars featuring both internal and external scholars. These seminars take place within the department as well as across the broader school.
- Student background
 - Students enter the program from a wide range of academic disciplines, including, but not limited to: economics, computer science, engineering, math, statistics, physics, psychology, and business
 - A master's degree is not required, although many students enter the program with one.
 - Prior research experience is highly beneficial but is not required for admission.
- Doctoral students typically receive financial support that includes:
 - Full tuition remission
 - Stipend and fellowships
 - Health insurance
 - Support for research activities and conference travel